

JEE ADVANCED TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

Sec: In Coming Sr. Super Chaina-R
Time: 2:00 PM to 05:00 PM

Date: 03-05-2020
Max. Marks: 186

Name of the Student: _____

H.T. NO:

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2016_P-2

03-05-2020_In Coming Sr. Super Chaina-R_Jee-Adv_CAT-03_SYLLABUS

PHYSICS : CUMULATIVE

CHEMISTRY : CUMULATIVE

MATHEAMTICS : CUMULATIVE

JEE-ADVANCE-2016-P2-Model

Time: 3-Hrs

IMPORTANT INSTRUCTIONS

Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 14)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 15 – 18)	Questions with ComprehensionType (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 32)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 33 – 36)	Questions with ComprehensionType (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

MATHS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 50)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 51 – 54)	Questions with ComprehensionType (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

PHYSICS

Max.Marks: 62

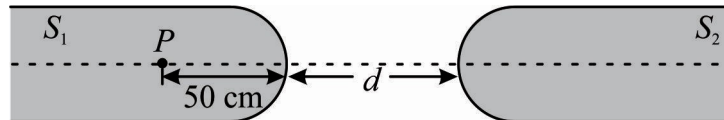
SECTION-I

(Single Correct Answer Type)

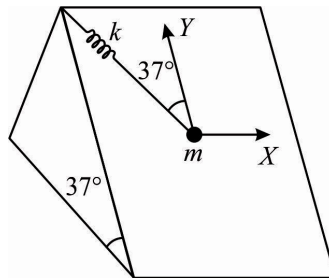
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

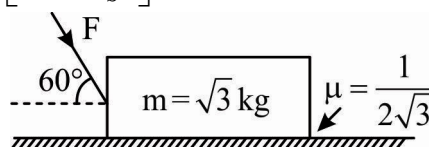
1. Two identical glass rods S_1 and S_2 (refractive index = 1.5) have one convex end of radius of curvature 10cm. They are placed with the curved surfaces at a distance d as shown in the figure, with their axes (shown by the dashed line) aligned. When a point source of light P is placed inside rod S_1 on its axis at distance of 50 cm from the curved face, the light rays emanating from it are found to be parallel to the axis inside S_2 . The distance d is



- (A) 60cm (B) 70 cm (C) 80cm (D) 90cm
2. The acceleration of the block of mass $m = 2$ kg is along negative X-axis (as shown in the figure). The magnitude of acceleration of the block is [where K =spring constant, $g = 10\text{ms}^{-2}$]

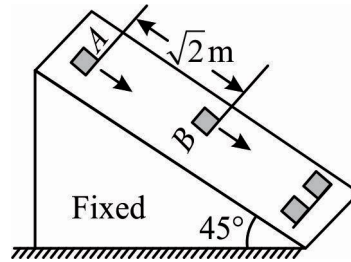


- (A) 1m/s^2 (B) 9m/s^2 (C) 4.5m/s^2 (D) 7.5m/s^2
3. What is the magnitude of maximum value of the force F such that the block shown in the arrangement, does not move? $\left[g = 10 \frac{\text{m}}{\text{s}^2} \right]$



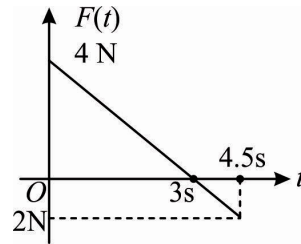
- (A) 20N (B) 10N (C) 12N (D) 15N
4. Two blocks A and B of equal mass are sliding down along straight parallel lines on an inclined plane of 45° . Their coefficients of kinetic friction are $\mu_A = 0.2$ and $\mu_B = 0.3$ respectively. At $t = 0$, both blocks are at rest and block A is $\sqrt{2}$ m behind block B. Find the time and distance from the

initial position of A, where the front faces of the blocks come in line on the inclined plane as shown in the figure (use $g = 10\text{ms}^{-2}$)



- (A) $2s, 8\sqrt{2}m$ (B) $\sqrt{2}s, 7m$ (C) $\sqrt{2}s, 7\sqrt{2}m$ (D) $2s, 7/\sqrt{2}m$

5. A block of mass 2kg is free to move along the x-axis. It is at rest and from $t = 0$ onwards it is subjected to a time-dependent force $F(t)$ in the x-direction. The force $F(t)$ varies with t as shown in the figure. The kinetic energy of the block after 4.5 s is



- (A) 4.50J (B) 7.50J (C) 5.06J (D) 14.06J

6. The work done on a particle of mass m by a force $k \left[\frac{X}{(X^2 + Y^2)^{3/2}} \hat{i} + \frac{Y}{(X^2 + Y^2)^{3/2}} \hat{j} \right] N$ (k is a constant of appropriate dimensions), When the particle is taken from the point $(a,0)$ to the point $(0,a)$ along a circular path of radius ' a ' about the origin in the XY-Plane is

- (A) $\frac{2k\pi}{a}$ (B) $\frac{k\pi}{a}$ (C) $\frac{k\pi}{2a}$ (D) zero

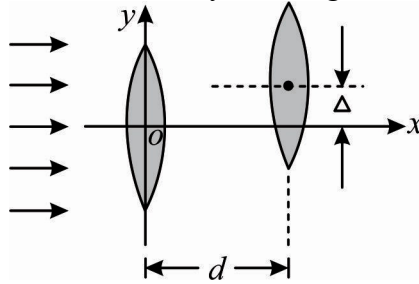
SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

7. Two thin convex lenses of focal lengths f_1 and f_2 are separated by a horizontal distance d (where $d < f_1, d < f_2$) and their centres are displaced by a vertical separation Δ as shown in the figure

Taking the origin of coordinates O at the centre of the first lens, the x and y coordinates of the focal point of this lens system, for a parallel beam of rays coming from the left, are given by



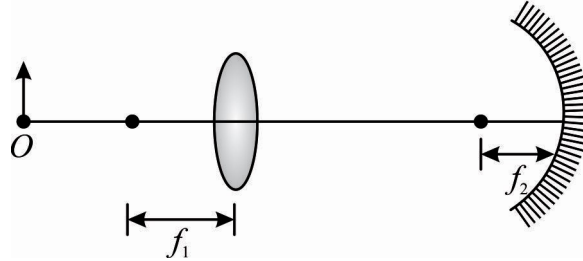
(A) $x = \frac{f_1 f_2}{f_1 + f_2}, y = \Delta$

(B) $x = \frac{f_1 f_2}{f_1 + f_2}, y = \frac{\Delta(f_1 - d)}{f_1 + f_2 - d}$

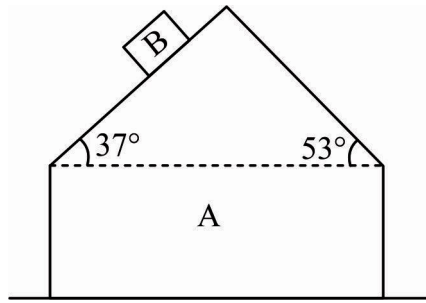
(C) $x = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}, y = \frac{\Delta(f_1 - d)}{f_1 + f_2 - d}$

(D) $x = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}, y = \Delta$

8. An object is placed in front of a converging lens at a distance equal to twice the focal length f_1 of the lens. On the other side of the lens is a concave mirror of focal length f_2 separated from the lens by a distance $2(f_1 + f_2)$. Light from the object passes rightward through the lens, reflects from the mirror, passes leftward through the lens, and forms a final image of the object

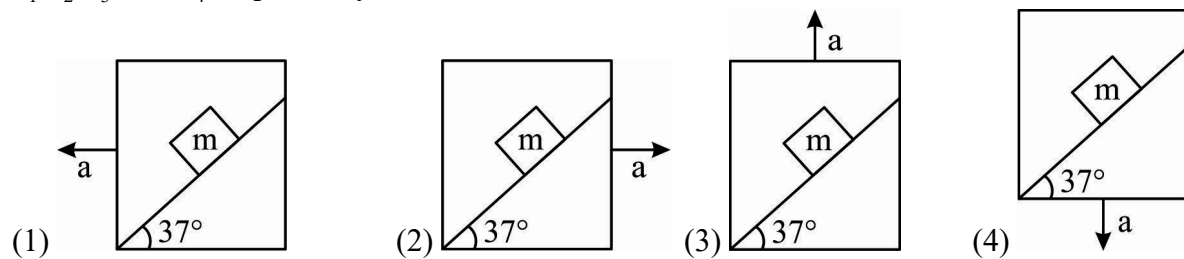


- (A) The distance between the lens and the final image is equal to $2f_1$
 (B) The distance between the lens and the final image is equal to $2(f_1 + f_2)$
 (C) The final image is real, inverted and of same size as that of the object.
 (D) The final image is real, erect and of same size as that of the object
9. Block A of mass 1 kg is placed on a smooth horizontal surface. A smooth block B of mass 1 kg starts to move on the inclined surface of block A . Then, $\left[\because g = 10 \frac{\text{m}}{\text{s}^2} \right]$

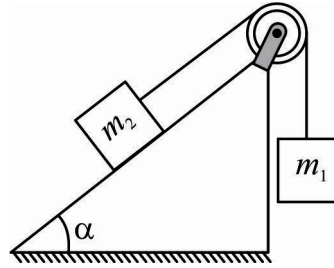


- (A) The acceleration of block A is $\frac{6g}{17}$
- (B) The acceleration of block B with respect to block A is $\frac{15g}{17}$
- (C) If acceleration of block B makes θ angle with horizontal, then $|\tan \theta| = 1.5$
- (D) The acceleration of block A is $\frac{g}{17}$

10. A block of mass m is placed on a wedge. The wedge can be accelerated in four manners marked as (1), (2), (3) and (4) as shown. If the normal reactions in situation (1), (2), (3) and (4) are N_1, N_2, N_3 and N_4 respectively and acceleration with which the block slides on the wedge in situation are b_1, b_2, b_3 and b_4 respectively then

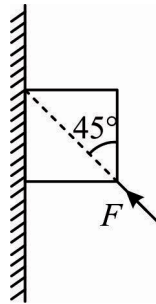


- (A) $N_3 > N_1 > N_2 > N_4$
- (B) $N_4 > N_3 > N_1 > N_2$
- (C) $b_2 > b_3 > b_4 > b_1$
- (D) $b_2 > b_3 > b_1 > b_4$
11. In the situation shown in the figure, the coefficients of static friction and kinetic friction are μ_s and μ_k respectively. The blocks are released from rest. Which of the following statement(s) is/are correct:

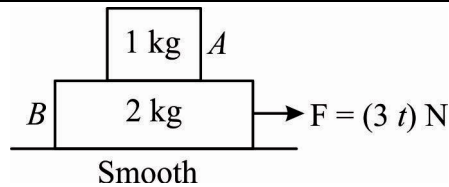


- (A) If $m_1g - m_2g \sin \alpha \leq \mu_s m_2g \cos \alpha$, the tension in string is m_1g and acceleration of blocks is zero
- (B) If $m_1g - m_2g \sin \alpha \leq \mu_s m_2g \cos \alpha$, the friction on m_2 is $(m_1g - m_2g \sin \alpha)$ in downward direction along the inclined plane
- (C) If $m_2g \sin \alpha - m_1g \leq \mu_s m_2g \cos \alpha$, the friction on m_2 is $(m_2g \sin \alpha - m_1g)$ in upward direction along inclined plane
- (D) If $m_1g - m_2g \sin \alpha > \mu_s m_2g \cos \alpha$, the acceleration of mass m_2 is $\left(\frac{m_1g - \mu_k m_2g \cos \alpha - m_2g \sin \alpha}{m_1 + m_2} \right)$ is upward direction along inclined plane

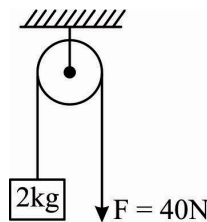
12. A block of mass 5kg is pressed against a vertical wall. The coefficients of static friction and kinetic friction between blocks are 0.5 and 0.4 respectively. Choose the correct statement(s) $\left[\because g = 10 \frac{m}{s^2} \right]$



- (A) If $F = 50\sqrt{2}N$, then friction on the block is zero
- (B) If $F = 40\sqrt{2}N$, then friction on the block is 10N in upward direction
- (C) If $F = 20\sqrt{2}N$, then acceleration of the block is $4.4 m/s^2$ in downward direction
- (D) If $F = 120\sqrt{2}N$, then acceleration of the block is $4.4 m/s^2$ in upward direction
13. A variable force $F = 3t N$ is applied on lower block B. The coefficient of friction between blocks is $\mu = 0.5$. Which of the following statement(s) is/are correct. $[g = 10ms^{-2}]$



- (A) For $0 \leq t \leq 5$ s, both blocks move with a common acceleration
 (B) For $t = 1$ s, the acceleration of blocks is 1 m/s^2 and friction on block A is 1 N in rightward direction
 (C) For $t = 10$ s, the acceleration of block B is 12.5 m/s^2
 (D) For $t = 10$ s, the acceleration of block A is 5 m/s^2
14. A block of mass 2 kg is hanging over a smooth and light pulley through a light and very long string. The other end of the string is pulled by a constant force $F = 40 \text{ N}$. At $t = 0$ the system is at rest as shown. Then in the time interval from $t = 0$ to $t = \frac{2}{\sqrt{10}}$ seconds, pick up the correct statement(s) : ($g = 10 \text{ m/s}^2$)



- (A) tension in the string is 40 N
 (B) work done by gravity is -20 J
 (C) work done by tension on block is 80 J
 (D) Velocity of the block at $t = \frac{2}{\sqrt{10}} \text{ s}$ is 30 m/s

SECTION-III

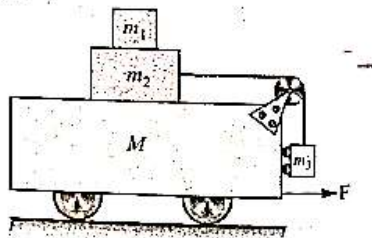
(Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 15 and 16

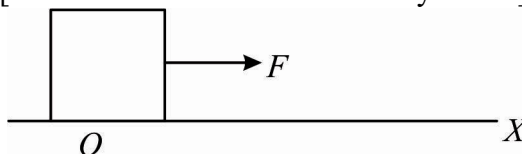
In the system shown in figure, $M = 13.4 \text{ kg}$, $m_1 = 1 \text{ kg}$, $m_2 = 2 \text{ kg}$ and $m_3 = 3.6 \text{ kg}$. The coefficient of static friction between m_1 and m_2 is 0.75 and that between m_2 and M is 0.6. All other surfaces are frictionless [$\because g = 9.8 \text{ m/s}^2$]



15. The minimum horizontal force F must be applied to M so that m_2 does not slip on M is
 (A) 117.7 N (B) 120.5 N (C) 125 N (D) 145.5 N
16. The maximum value of F for which m_1 does not slip on m_2
 (A) 125 N (B) 147 N (C) 165 N (D) 175 N

Paragraph For Questions 17 and 18

A block of mass m is initially at $x = 0$ is pulled along X -axis by applying the force $F = b - cx$, where b and c are positive constant(s) [Assume that the block is initially at rest]



17. The maximum value of x is
 (A) $\frac{b}{c}$ (B) $\frac{b}{2c}$ (C) $\frac{2b}{c}$ (D) $\frac{b}{3c}$
18. The maximum speed of the block during motion of the block is
 (A) $\frac{b}{\sqrt{mc}}$ (B) $\frac{2b}{\sqrt{mc}}$ (C) $\frac{3b}{\sqrt{mc}}$ (D) $\frac{4b}{\sqrt{mc}}$

CHEMISTRY

Max.Marks: 62

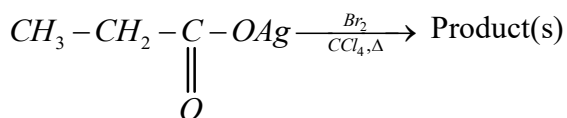
SECTION-I

(Single Correct Answer Type)

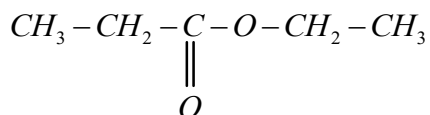
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

19. Which can not be the possible product of the given reaction?



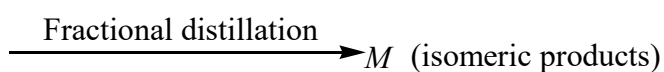
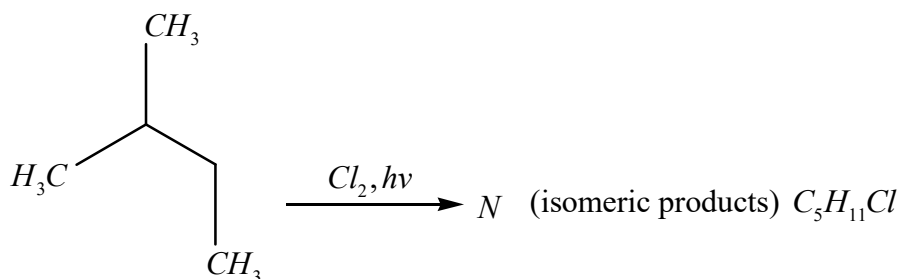
- (A) $\text{CH}_3 - \text{CH}_2 - \text{Br}$



(B)

(C) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$ (D) $\text{CH}_3-\text{CH}_2-\text{CH}_3$

20.



What are N and M?

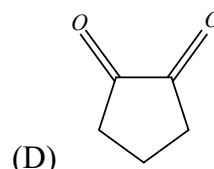
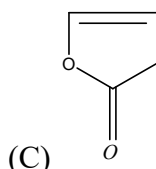
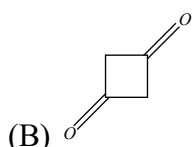
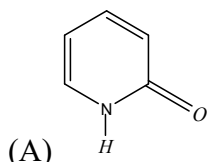
(A) 6,6

(B) 6,4

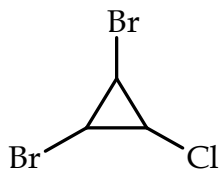
(C) 4,4

(D) 3,3

21. Which of the following has the least stable enol form?



22. No of stereoisomers possible for the following compound is



(A) 2

(B) 3

(C) 4

(D) 6

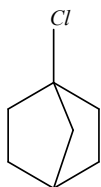
23. The $\text{S}_{\text{N}}2$ reaction of 1-chloro-3-methylcyclobutane with sodium methoxide is relatively slow, but can be accelerated by the addition of a small amount NaI. How is this catalysis best explained?

(A) The sodium cation helps pull off the chloride

(B) The iodide anion activates the methoxide nucleophile

(C) $\text{S}_{\text{N}}2$ reaction of iodide ion converts the alkyl chloride to the more reactive alkyl iodide(D) The NaI changes the mechanism to $\text{S}_{\text{N}}1$

24. Which of the following will give white ppt. with $aq. AgNO_3$?



- (A) $Ph-Cl$ (B) $H_2N-CH_2CH_2CH_2CH_2-Cl$
 (C) $Ph-CH_2-I$

SECTION-II

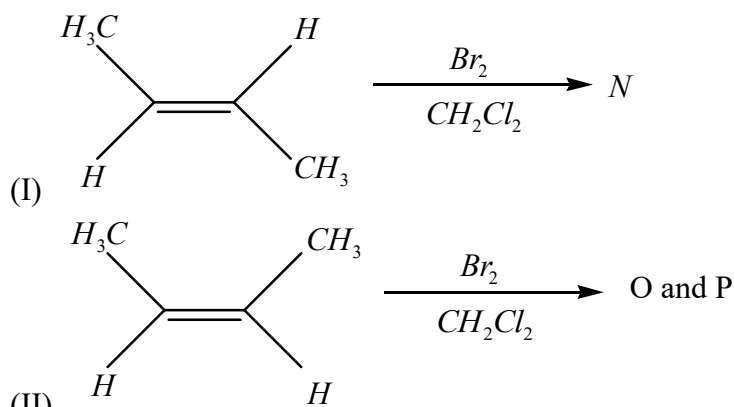
(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

25. Which orders are correct?

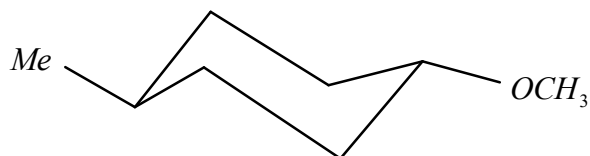
- (A) Thermal stability: $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$
 (B) Basic nature: $BeO > MgO > CaO > SrO$
 (C) Solubility in water: $LiOH > NaOH > KOH > RbOH > CsOH$
 (D) Melting point: $NaCl > KCl > RbCl > CsCl > LiCl$

26. The correct statement(s) for the following addition reactions is(are)

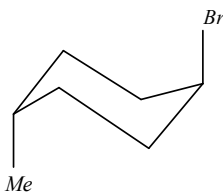


- (A) O and P are identical molecules
 (B) Bromination proceeds through *trans* addition in both the reactions
 (C) (N and O) and (N and P) are two pairs of enantiomers.
 (D) (N and O) and (N and P) are two pairs of diastereomers.

- 27.



can be synthesised from:

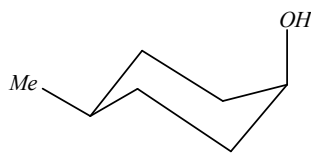


(A) Reaction of CH_3OH with K followed by



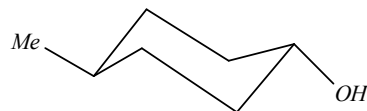
(B) Reaction of

with SOCl_2 followed by $\text{CH}_3\text{O}^\ominus\text{Na}^\oplus$



(C) Reaction of

with K followed by CH_3Cl

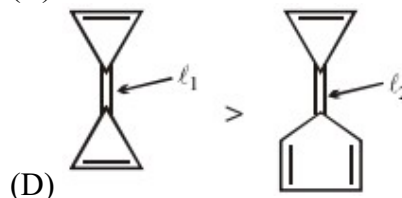
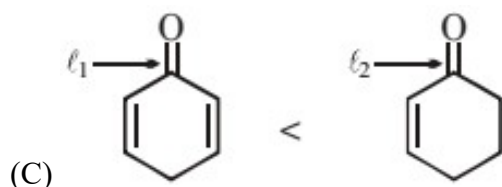
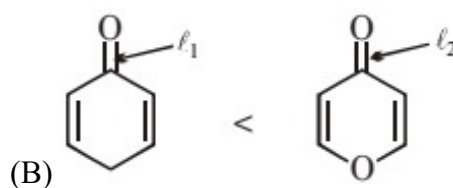
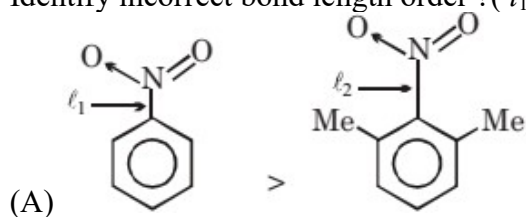


(D) Reaction of

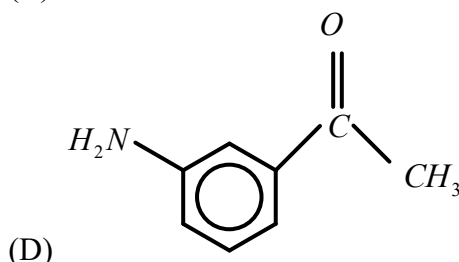
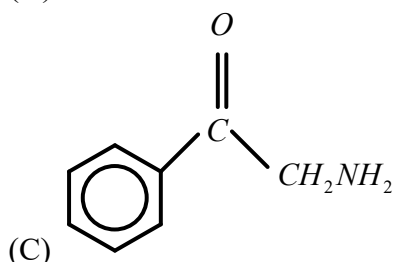
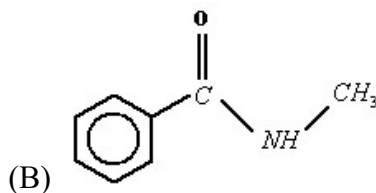
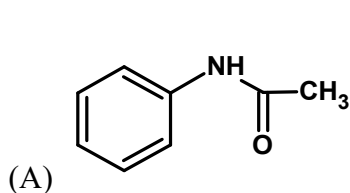
with SOCl_2 in presence of pyridine followed by

$\text{CH}_3\text{O}^\ominus\text{Na}^\oplus$

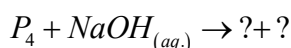
28. Identify incorrect bond length order ? (l_1 & l_2)



29. Which is less basic than benzyl amine



30. Select correct statements about following reaction:



- (A) One of the products is PH_3
 (B) It is disproportionation reaction
 (C) One of the products is NaH_2PO_3
 (D) This reaction is not given by red phosphorous (Room temperature)
31. Choose the correct statement(s):
 (A) $BeCO_3$ is kept in the atmosphere of CO_2 since, it is least thermally stable
 (B) Be dissolves in an alkali solution forming $[Be(OH)_4]^{2-}$
 (C) Sodium metal in $liq. NH_3$ solution is diamagnetic in nature (1 M)
 (D) BeF_2 forms complex ion with NaF in which Be goes with anion
32. In which of the following sets of reagents, all the reagents react with caustic soda (NaOH) to give a gaseous product.
 (A) P_4, F_2, NH_4Cl (B) B, Al, Si (C) Zn, Sn, Pb (D) $S_8, Cl_2, ZnSO_4$

SECTION-III

(Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

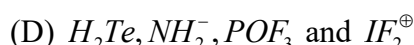
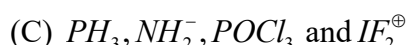
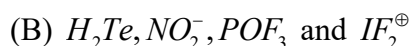
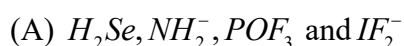
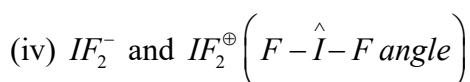
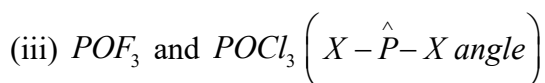
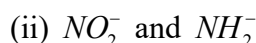
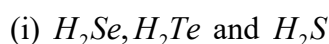
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 33 and 34

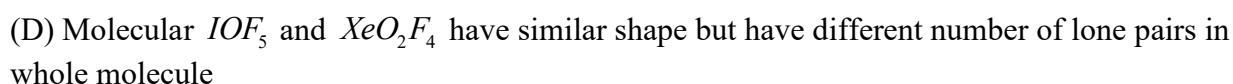
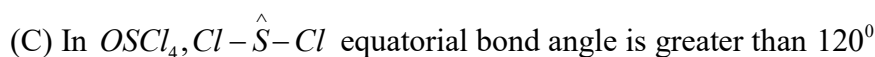
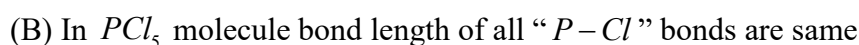
The concept of redistribution of energy in different orbitals of an atom associated with different energies to give new orbitals of equal (or sometimes it may be non-equal) energy oriented in space in

definite directions is called hybridization and formed new orbitals are called hybrid orbitals. The bonds formed by such orbitals are called hybrid bonds. The process of mixing of orbitals itself requires some energy. Thus, some additional energy, is needed for the hybridization (mixing) of atomic orbitals.

33. Select from each set the molecule or ion having the smallest bond angle:



34. Which of the following statement is correct?



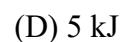
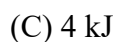
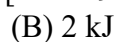
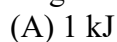
Paragraph For Questions 35 and 36

The minimum amount of energy required to remove the most loosely bound electron from an isolated atom in the gaseous state is known as ionization energy or first ionization energy or ionization enthalpy (IE_1) of the element.

The energy required to remove the second electron from the monovalent cation is called second ionization enthalpy (IE_2). Similarly, we have third, fourth.....ionisation enthalpies. The values of ionization energy depends on a number of factors such as (i) size of the atom (ii) screening effect (iii) nuclear charge (iv) half filled and fully filled orbitals (v) shape of orbital.

In a group, the ionization energy decreases from top to bottom. In a period, the value of ionization energy increases from left to right with breaks where atoms have somewhat stable configurations.

35. First and second ionization energies of magnesium are 5.7 and 15.035 eV respectively. The amount of energy in kJ needed to convert all the atoms of magnesium into Mg^{+2} ions present in 12 mg of magnesium vapour is [$1\text{eV} = 96.5\text{kJmol}^{-1}$]



36. First and second Ionisation energies of Mg are 740 and 1450 kJ/mol respectively. If 1 g of $Mg(g)$ absorbed 50 kJ energy then the no. of moles of Mg^+ in the mixture formed is
- (A) 0.0132 (B) 0.0285 (C) 0.0715 (D) 0.0632

SECTION-I

(Single Correct Answer Type)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. $\lim_{x \rightarrow 0} \left(\frac{(1+x)^{\frac{1}{x}}}{e} \right)^{\frac{1}{x}} =$

- (A) 1 (B) $\sqrt{e}$ (C) $\frac{1}{\sqrt{e}}$ (D) $2e$

38. If $\lim_{x \rightarrow 0^+} \frac{(x^3 - \sin^3 x) \log_e (1+3x)}{\left(\left[\frac{\tan x}{x} \right] - \cos 3x \right) \cdot x^n} = l$ where l is finite and non-zero then $(n+l)$ equals (Where $[.]$

denotes greatest integer function)

- (A) $\frac{13}{3}$ (B) $\frac{11}{3}$ (C) $\frac{17}{3}$ (D) $\frac{19}{3}$

39. If the equation $\left| x-1 \right| - 6 \lim_{t \rightarrow \infty} \left(\frac{\sqrt{2t^2 - t - 1} - \sqrt{t^2 - t + 1}}{t \left(\tan \frac{\pi}{8} \right)} \right) = 2k$ has four distinct solutions then the

number of integral values of k is,

- (A) 4 (B) 3 (C) 2 (D) 1

40. If $L = \lim_{x \rightarrow 0} \frac{p^2 x + (p-2) \sin x}{(\tan^{-1} x)^3}$, where $p \in R$, is finite then

- (A) $p = 3$ (B) $p = -5$ (C) $L = \frac{1}{6}$ (D) $L = 7$

41. Let $\langle a_m \rangle$ be the m^{th} term of the sequence, $\left(\frac{2^2}{1^2} - \frac{2}{1} \right)^{-1}, \left(\frac{3^3}{2^3} - \frac{3}{2} \right)^{-2}, \left(\frac{4^4}{3^4} - \frac{4}{3} \right)^{-3}, \dots$

The value of $\lim_{m \rightarrow \infty} (a_m)^{1/m}$ is equal to:

- (A) $e+1$ (B) $e-1$ (C) $\frac{1}{e-1}$ (D) $\frac{1}{e+1}$

42. If the circle $S_1 : x^2 + y^2 = 16$ intersects another circle S_2 of radius 5 in such a manner that the common chord is of maximum length and has a slope equal to $\frac{3}{4}$, the coordinates of the centre of S_2 are
- (A) $\left(-\frac{9}{5}, \frac{12}{5}\right), \left(\frac{9}{5}, -\frac{12}{5}\right)$ (B) $\left(-\frac{9}{5}, -\frac{12}{5}\right), \left(\frac{9}{5}, \frac{12}{5}\right)$
- (C) $\left(\frac{12}{5}, \frac{9}{5}\right), \left(-\frac{12}{5}, \frac{9}{5}\right)$ (D) $\left(\frac{12}{5}, -\frac{9}{5}\right), \left(-\frac{12}{5}, -\frac{9}{5}\right)$

SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

43. Let A, B and C be three distinct points on $y^2 = 8x$ such that normals at these points are concurrent at P . The slope of AB is 2 and abscissa of centroid of $\triangle ABC$ is $\frac{4}{3}$. Which of the following is TRUE.
- (A) Area of $\triangle ABC$ is 8 sq.units (B) Coordinates of $P = (6, 0)$
- (C) Circumcentre of $\triangle ABC$ is $(2, 0)$ (D) Angle between normals are $45^\circ, 45^\circ, 90^\circ$
44. The tangent PT and normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N respectively. The locus of the centroid of the triangle PTN is a parabola whose $(a > 0)$
- A) Vertex is $\left(\frac{2a}{3}, 0\right)$ B) focus is $(2a, 0)$
- C) Length of latus rectum is $\frac{4a}{3}$ D) directrix is $x = \frac{a}{3}$
45. If $f(x) = \lim_{p \rightarrow \infty} \frac{x^p g(x) + h(x) + 7}{7x^p + 3x + 1}$; $x \neq 1$ and $f(1) = 7$, $f(x)$, $g(x)$ and $h(x)$ are all continuous functions at $x = 1$. Then which of the following statement(s) is/are correct

- (A) $g(1) + h(1) = 70$ (B) $g(1) - h(1) = 28$
 (C) $g(1) + h(1) = 60$ (D) $g(1) - h(1) = -28$
46. If two distinct chords of a parabola $y^2 = 4ax$ passing through the point $(a, 2a)$ are bisected by line $x + y = 1$, Then the length of the latusrectum cannot be:
 (A) 2 (B) 4 (C) 5 (D) 7
47. Let $f(x) = \frac{1 - \cos\{x\}}{(x^4 + ax^3 + bx^2 + cx)^2}$. If $l = \lim_{x \rightarrow 1^+} f(x)$, $m = \lim_{x \rightarrow 2^+} f(x)$ and $n = \lim_{x \rightarrow 3^+} f(x)$, where l, m and n are non-zero finite then:
 (A) $\lim_{x \rightarrow 0^+} f(x) = \frac{1}{36}$ (B) $a + b + c = -1$
 (C) $l + m + n = \frac{19}{72}$ (D) $\lim_{x \rightarrow 4^+} f(x) = 0$
- [Note: $\{y\}$ denotes fractional part of function of 'y'.]
48. If $f(x) = \log \left\{ \log_{|\sin x|} (x^2 - 8x + 23) - \frac{3}{\log_2 |\sin x|} \right\}$, then which of the following is a subset of the domain of $f(x)$?
 (A) $\left(0, \frac{\pi}{2}\right)$ (B) $(3, \pi)$ (C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) $\left(\frac{3\pi}{2}, 5\right)$
49. If $2f(x) + xf\left(\frac{1}{x}\right) - 2f\left(\left|\sqrt{2} \sin \pi \left(x + \frac{1}{4}\right)\right|\right) = 4\cos^2\left(\frac{\pi x}{2}\right) + x \cos \frac{\pi}{x}, \forall x \in R - \{0\}$ then which of the following statement(s) is/are true?
 (A) $f(2) + f\left(\frac{1}{2}\right) = 1$ (B) $f(2) + f(1) = 0$
 (C) $f(2) + f(1) = f\left(\frac{1}{2}\right)$ (D) $f(1)f\left(\frac{1}{2}\right)f(2) = 1$
50. Let $f: N \times N \rightarrow N$ be a function such that $f(1,1) = 2$ &
 $f(\alpha + 1, \beta) = f(\alpha, \beta) + \alpha$ & $f(\alpha, \beta + 1) = f(\alpha, \beta) - \beta \forall \alpha, \beta \in N$ & $f(a, b) = 2001, a, b \in N$ then which of the following statement(s) is/are true?
 (A) Number of ordered pairs (a, b) is 2 (B) Maximum value of $a + b$ is 3999
 (C) Minimum value of $a + b$ is 2000 (D) Minimum value of $a + b$ is 1999

SECTION-III**(Paragraph Type)**

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 51 and 52

Consider two circles $S_1 = 0$ and $S_2 = 0$, each of radius 1 unit touching internally the sides of $\triangle OAB$ and $\triangle ABC$ respectively where $O = (0, 0)$, $A(0, 4)$ and B, C are points on positive x -axis such that $OB < OC$.

51. If the angle subtended by the circle $S_1 = 0$ at point A is θ , then $\cos \theta$ equals

- (A) $\frac{3}{5}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$

52. The length of tangent from A to the circle $S_2 = 0$, is

- (A) $\frac{5}{2}$ (B) $\frac{7}{2}$ (C) $\frac{9}{2}$ (D) $\frac{11}{2}$

Paragraph For Questions 53 and 54

Let $P(x)$ be a polynomial of degree '6' with leading coefficient unity and $P(-x) = P(x) \forall x \in \mathbb{R}$.

$$\text{Also } (P(1)+3)^2 + P^2(2) + (P(3)-5)^2 = 0$$

53. The value of $\lim_{x \rightarrow -2} \frac{\sin(P(x))}{(x-2)\tan(x+2)}$ is equal to:

- (A) 26 (B) 16 (C) -14 (D) -40

54. The value of $\lim_{x \rightarrow \infty} \frac{\left(x^2 - \sqrt{\frac{P(x)}{x^2 - 4}}\right)}{x \tan \frac{1}{x}}$ is equal to:

- (A) 10 (B) 5 (C) 3 (D) -4